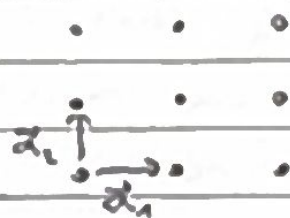


1. Primitivna rešetka, primitivni vektorji

$\vec{a}_1, \vec{a}_2, \vec{a}_3$ primitivni vektorji
(najmanjši možni)



$$\vec{R} = n_1 \vec{a}_1 + n_2 \vec{a}_2 + n_3 \vec{a}_3, n_i \in \mathbb{Z}$$

če se premaknemo za $\vec{R}$ v prostoru, je okolica enaka kot prej

Primitivna rešetka (celica) je najmanjša prostorska enota kristalne rešetke, ki z zaporednimi translacijami vzdolž primitivnih vektorjev zapolni celoten prostor brez vrzeli/prekrivanja. V primitivni celici je en atom in neničelno določena.

Konvencionalna celica ima lahko v sebi več atomov in je izbrana tako, da kaže polno geometrijsko simetrijo (npr. ploskavno cent. kaka)

2. Simetrijske operacije na kristalu

Translacija: $\hat{T} = n_1 \vec{a}_1 + n_2 \vec{a}_2 + n_3 \vec{a}_3, \quad \hat{T} \psi(\vec{r}) = \psi(\vec{r} - \vec{R})$

Točkovne simetrije: rotacija: C_n, n -števna os: $n=2, 3, 4, 6, \varphi = \frac{360^\circ}{n}$

zrcaljenja:

Inverzija: $\vec{R} \rightarrow -\vec{R}$

rotoinverzija: inverzija + rotacija

operacija lahko zapisano z matriko, $\det = \pm 1$

(točkovne)

operacije tvorijo točkovno simetrijsko grupo ($g \in G$):

- $\exists E$ (identiteta, npr. C_1 - zasuk za $4 \cdot 90^\circ$ okoli 4-števne osi)

- $\forall g \exists g^{-1}$

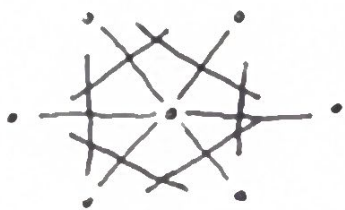
- $\forall g_1, g_2 \Rightarrow g_1 g_2 \in G$

- $g_1 (g_2 g_3) = (g_1 g_2) g_3$

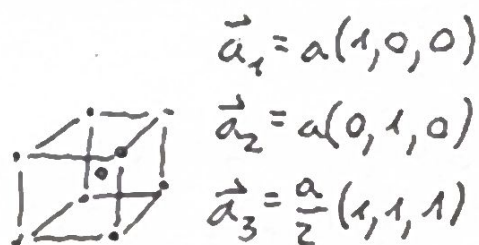
3. Primeri in opisi najbolj značilnih Brav. rešetk

• Wigner-Seitzova osn. celica

osn. celico dobimo tako, da na polovični razdalji med najbl. sosedi nekega delca narišemo pravokotno ravnino



• Telesno centrirana kubična



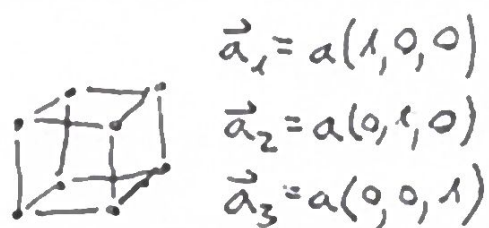
$$N_c = 8$$

• Diamantna rešetka

$$FCC + \vec{a}_1 = 0, \vec{a}_2 = \frac{a}{4}(1,1,1)$$

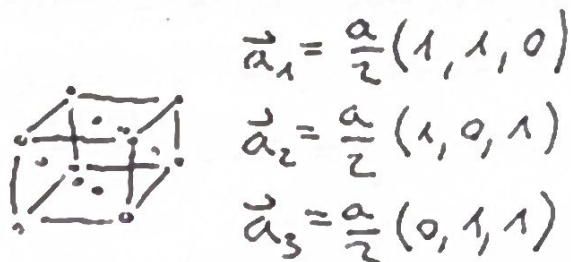
$$N_c = 4$$

• Preprosta kubična



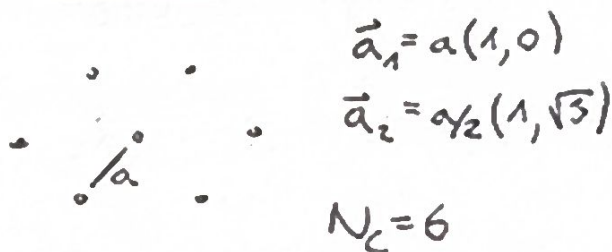
$N_c = 6$ Koordinacijsko število - kolikoli najbl. sosedov ima atom/delec

• Ploskovno centrirana kubična



$$N_c = 12$$

• Trikotna mreža (2D)



$$N_c = 6$$

4. Recipročna rešetka

$$e^{i\vec{k} \cdot (\vec{r} + \vec{R})} = e^{i\vec{k} \cdot \vec{r}}$$

$$\rightarrow e^{i\vec{k} \cdot \vec{R}} = 1 \quad \forall \vec{R} \in \text{Brav. reš.}$$

$$\vec{k} = ?$$

$$\vec{A}_1 = \frac{\vec{a}_2 \times \vec{a}_3}{(\vec{a}_1, \vec{a}_2, \vec{a}_3)} 2\pi$$

$$\vec{A}_2 = \frac{\vec{a}_3 \times \vec{a}_1}{(\vec{a}_1, \vec{a}_2, \vec{a}_3)} 2\pi$$

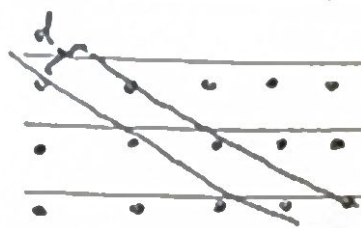
$$\vec{A}_3 = \frac{\vec{a}_1 \times \vec{a}_2}{(\vec{a}_1, \vec{a}_2, \vec{a}_3)} 2\pi$$

$$\vec{a}_i, \vec{A}_j = 2\pi \delta_{ij} \Rightarrow \vec{R} \cdot \vec{A}_i = 2\pi n_i$$

$$\rightarrow \vec{k} = k_1 \vec{A}_1 + k_2 \vec{A}_2 + k_3 \vec{A}_3, \quad k_i \in \mathbb{Z}$$

Recipročna rešetka je matematični konstrukt, ki predstavlja periodičnost kristala v prostoru valovnih vektorjev $\vec{k}$.

5. Mrežne ravnine in Millerjevi indeksi



$$\vec{K} = \frac{2\pi}{\lambda} \hat{n}$$

$$\vec{R} \cdot \vec{K} = c \text{ (ravnina)}$$

$$\vec{R}_1 \cdot \vec{K} = c_1, \vec{R}_2 \cdot \vec{K} = c_2 \text{ vektorja iz sosednjih ravnin}$$

$$(\vec{R}_2 - \vec{R}_1) \cdot \vec{K} = c_2 - c_1 = \underbrace{(\vec{R}_2 - \vec{R}_1) \cdot \hat{n}}_d \frac{2\pi}{\lambda} = 2\pi$$

$$\rightarrow \hat{n} (\vec{R}_2 - \vec{R}_1) \frac{2\pi}{\lambda} = 2\pi k, k \in \mathbb{Z} \Rightarrow \vec{R}_1 \cdot \vec{K} = 2\pi k$$

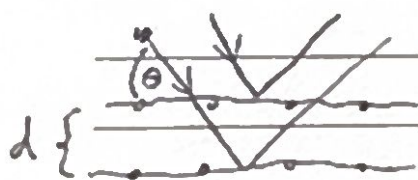
$$\text{Millerjevi indeksi } h, k, l : \vec{K} = h\vec{A}_1 + k\vec{A}_2 + l\vec{A}_3$$

$$x_1 \vec{a}_1 \cdot \vec{K} = x_1 h 2\pi, x_2 \vec{a}_2 \cdot \vec{K} = x_2 k 2\pi, x_3 \vec{a}_3 \cdot \vec{K} = x_3 l 2\pi$$

$$\Rightarrow \text{Razmerje } h:k:l = \frac{1}{x_1} : \frac{1}{x_2} : \frac{1}{x_3}$$

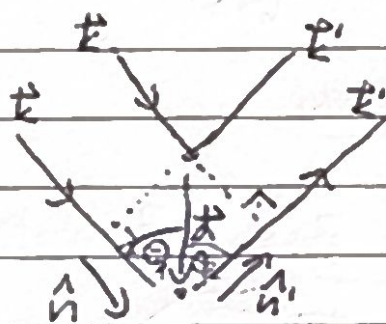
6. Braggovo sipanje in von Lauejeva formulacija sipanja

Bragg:



$$2d \sin \theta = n\lambda$$

Von Laue:



$$d \cos \theta_1 + d \cos \theta_2 = m\lambda$$

$$\vec{d} \cdot (\hat{n} - \hat{n}') = m\lambda / \cdot \frac{2\pi}{\lambda}$$

$$\vec{d} \cdot (\vec{k} - \vec{k}') = 2\pi m$$

$$\vec{k} - \vec{k}' = \vec{K}$$

$$\forall \vec{K} \in \text{Brav. res.}$$

7. Geometrijski strukturni faktor

$$S_{\vec{k}} = \sum_{j=1}^n e^{i\vec{k} \cdot \vec{d}_j} \quad \begin{array}{l} n \dots \text{št. atomov v osn. celici} \\ \vec{d}_j \dots \text{baza sipalca v osn. celici} \end{array}$$

Intenziteta $\propto |S_{\vec{k}}|^2$

$S_{\vec{k}}$ predstavlja vir $\vec{k}$ -odvisnosti Bragg. sip. Če $S_{\vec{k}}=0$, ta ravnina ne prispeva k sipanju

večatomni kristal

$$S_{\vec{k}} = \sum f_j(\vec{k}) e^{i\vec{k} \cdot \vec{d}_j}, \quad f_j(\vec{k}) = -\frac{1}{V} \int d\vec{r} e^{i\vec{k} \cdot \vec{r}} \rho_j(\vec{r})$$

$\hookrightarrow$ el. gostota

če $\vec{k} \cdot \vec{r} \ll 1$: $f_j(\vec{k}) \approx Z$

8. Klasifikacija Brav. rešetk

• 7 kristalnih sistemov:

- kubični ($\alpha=\beta=\gamma=90^\circ$, aaa)
- tetragonalni ($\alpha=\beta=\gamma=90^\circ$, aab)
- ortorombski ($\alpha=\beta=\gamma=90^\circ$, abc)
- monoklinski ($\alpha=\beta=90^\circ \neq \gamma$, abc)
- triklinski ($\alpha \neq \beta \neq \gamma \neq 90^\circ$, abc)
- trigonalni ($\alpha \neq \beta \neq \gamma \neq 90^\circ$, aaa)
- heksagonalni ($\alpha=120^\circ$, $\beta=60^\circ$, $\gamma=90^\circ$, aab)

Brav. reš.

3 (sc, fcc, bcc)

2 (sq, bca)

4 (sc, fcc, bcc, oglišča + 2 naspr. stranici)

2 (sq, bca)

1 (sc)

1 (sc)

1 (sc)

	baza sf. simetrije	baza arbitrarn. $\Sigma=14$ simetrije
točkovne grupe:	7 krist. sist.	32 točk. grup
prost. grupe	14 Brav. rešetk	230 prost. grup

Zakaj $\exists$ n-sterne rotacije le za $n=1,2,3,4,6$?



$$a - 2a \cos \varphi = ma \rightarrow \cos \varphi = \frac{1-m}{2}$$

Točkovne simetrijske operacije tvorijo grupo: $\{A, B, C\} \in G$

$- AB = C \in G$

$- \exists E \in G; AE = EA = A \forall A$ identiteta

$- \exists A^{-1} \in G; AA^{-1} = A^{-1}A = E$ inverz

$- A(BC) = (AB)C$

m	3	2	1	0	-1
$\cos \varphi$	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1
φ	180°	120°	90°	60°	0°
n	2	3	4	6	1

9. Model prostih e^-

$$\langle l \rangle \sim 1 \text{ nm} \gg a \Rightarrow V \approx 0$$

$$-\frac{\hbar^2}{2m} \nabla^2 \psi = E \psi, \psi = e^{\pm i \vec{k} \cdot \vec{r}} \Rightarrow \xi = \frac{\hbar^2 k^2}{2m}$$

Periodični RP: $\psi(x+l, y, z) = \psi(x, y, z)$, enako za y in z

$$e^{i(k_x(x+l) + k_y y + k_z z)} = e^{i(k_x x + k_y y + k_z z)} \Rightarrow e^{i k_x l} = 1 = k_x l = n_1 2\pi$$

$$\rightarrow \xi_F = \frac{\hbar^2}{2m} \left(\frac{2\pi}{l} \right)^2 (n_1^2 + n_2^2 + n_3^2)$$

Število stanj $N = 2 \frac{V_{\text{Ferm. krogle}}}{V_{\text{element v k-prost.}}}$ $= 2 \frac{4\pi k_F^3}{3 \left(\frac{2\pi}{l} \right)^3} \rightarrow k_F = \left(3\pi^2 \frac{N}{V} \right)^{1/3}$
spin

$$\rightarrow \xi_F = \frac{\hbar^2 k_F^2}{2m} = \frac{\hbar^2}{2m} \left(3\pi^2 n_{el} \right)^{2/3} \quad \frac{\hbar k_F}{m} = v_F \rightarrow v_F = \frac{\hbar}{m} \left(3\pi^2 n_{el} \right)^{1/3}$$

Gostota stanj: $N = \frac{V}{3\pi^2} \left(\frac{2m\xi}{\hbar^2} \right)^{3/2} \rightarrow D(\xi) \cdot \frac{dN}{d\xi} = \frac{V}{2\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \sqrt{\xi}$

$$g(\xi) = \frac{1}{V} \frac{dN}{d\xi} = \frac{1}{2\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \sqrt{\xi} = \frac{3}{2} \frac{n}{\xi_F} \sqrt{\frac{\xi}{\xi_F}}$$

10. Temperaturna odvisnost c_v in μ v modelu prostih elektronov

$$f(\epsilon) = \frac{1}{e^{(\epsilon - \mu)/k_B T} + 1}$$

$$T=0: f(\epsilon) = \begin{cases} 1; & \epsilon < \epsilon_F \\ 0; & \text{sicer} \end{cases}$$

$$\mu(T=0) = \epsilon_F$$

$$c_v = \frac{1}{V} \left(\frac{\partial U}{\partial T} \right)_V = \left(\frac{\partial u}{\partial T} \right)_V \text{ gostota not. en.}$$

$$u(T=0) = \frac{2}{V} \sum_{\epsilon < \epsilon_F} \epsilon \quad u(T) = \frac{2}{V} \sum_{\epsilon} \epsilon f(\epsilon) = \frac{1}{4\pi^3} \int f(\epsilon) \epsilon d^3k = \int_{-\infty}^{\infty} f(\epsilon) \epsilon g(\epsilon) d\epsilon$$

$$\frac{d^3k}{4\pi^3} = \frac{4\pi k^2 dk}{4\pi^3} = g(\epsilon) d\epsilon \rightarrow \frac{1}{g(\epsilon)} = \pi^2 \frac{d\epsilon}{k^2 dk} = 2\pi \left(\frac{\hbar^2}{2m} \right)^{3/2} \frac{1}{\sqrt{\epsilon}}$$

$$n(T) = \int \frac{d^3k}{4\pi^3} f(\epsilon) = \int_{-\infty}^{\infty} f(\epsilon) g(\epsilon) d\epsilon$$

Sommerfeldov razvoj: $\int_{-\infty}^{\mu} H(\epsilon) f(\epsilon) d\epsilon = \int_{-\infty}^{\mu} H(\epsilon) d\epsilon + \frac{\pi^2}{6} H'(\mu) (k_B T)^2 + o(T^4)$

$$u(T) = \int_{-\infty}^{\mu} d\epsilon \epsilon g(\epsilon) + \frac{\pi^2}{6} (k_B T)^2 [\mu g'(\mu) + g(\mu)], \quad n(T) = \int_{-\infty}^{\mu} g(\epsilon) d\epsilon + \frac{\pi^2}{6} (k_B T)^2 g'(\mu) =$$

$$= \underbrace{\int_{-\infty}^{\epsilon_F} g(\epsilon) d\epsilon}_{n(T=0)} + \underbrace{\int_{\epsilon_F}^{\mu} g(\epsilon) d\epsilon}_{\mu \approx \epsilon_F \Rightarrow (\mu - \epsilon_F) g(\epsilon_F)} + \frac{\pi^2}{6} (k_B T)^2 g'(\mu) \approx n(T=0) + (\mu - \epsilon_F) g(\epsilon_F) + \frac{\pi^2}{6} (k_B T)^2 g'(\epsilon_F)$$

$$V = \text{konst.} \Rightarrow n(T) = n(0)$$

$$\Rightarrow \mu = \epsilon_F - \frac{\pi^2}{6} (k_B T)^2 \frac{g'(\epsilon_F)}{g(\epsilon_F)} = \epsilon_F \left(1 - \frac{\pi^2}{12} \left(\frac{k_B T}{\epsilon_F} \right)^2 \right)$$

$$u = \underbrace{\int_{-\infty}^{\epsilon_F} g(\epsilon) \epsilon d\epsilon}_{u_0} + (\mu - \epsilon_F) \epsilon_F g(\epsilon_F) + \frac{\pi^2}{6} (k_B T)^2 (g(\epsilon_F) + \epsilon_F g'(\epsilon_F)) = u_0 + \frac{\pi^2}{6} (k_B T)^2 g(\epsilon_F)$$

$$\rightarrow c_v(T) = \frac{\pi^2}{3} k_B T g(\epsilon_F) k_B = \frac{\pi^2}{2} \frac{k_B T}{\epsilon_F} n k_B$$

11. Drudejev model el. prevodnosti

Predpostavke:

- Z elektronov na atom je prostih, $Z_A - Z$ so vezani
- med trki so e^- prosti, trki so kratki
- verjetnost za trk na čas. enoto $1/\tau$, neodv. od $\vec{r}, \vec{v}$
- e^- vzpostavijo termično ravnovesje izključno prek trkov z ioni, ki so v term. rav. z okolico. Hitrost in smer gibanja po trku je naključna.

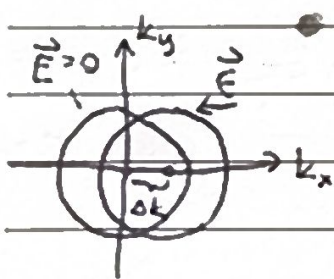
$$\vec{E} = \frac{1}{\sigma} \vec{j} \rightarrow \text{gostota el. toka}$$

↳ el. upornost

$$\vec{j} = -ne\vec{v}$$

$$m\vec{a} = -e\vec{E} - \frac{m\vec{v}}{\tau} \rightarrow \vec{v} = -\frac{e\vec{E}\tau}{m}, \quad \vec{j} = \frac{ne^2\tau}{m} \vec{E} \Rightarrow \sigma = \frac{ne^2\tau}{m}$$

↳ relaksacijski čas



$$\vec{F} = m\vec{a} = \hbar \frac{d\vec{k}}{dt} = -e\vec{E} \Rightarrow \Delta \vec{k} = -\frac{e\vec{E}\tau}{\hbar}$$

$$l = \tau v_F \text{ povpr. prosti pot}, \quad v_F = \frac{\hbar k_F}{m}$$

12. Elektroni v periodičnem potencialu, Blochov teorem

$$U(\vec{r} + \vec{R}) = U(\vec{r}), \quad \vec{R} \in \text{Brav. res.} \quad \left[-\frac{\hbar^2}{2m} \nabla^2 + U(\vec{r}) \right] \Psi = \epsilon \Psi$$

Blochov teorem: če velja $U(\vec{r} + \vec{R}) = U(\vec{r}) \Rightarrow \Psi_{n\vec{k}}(\vec{r}) = e^{i\vec{k}\vec{r}} u_{n\vec{k}}(\vec{r})$,

$$u_{n\vec{k}}(\vec{r} + \vec{R}) = u_{n\vec{k}}(\vec{r}). \quad \forall \vec{k} \exists \text{ več rešitev} \Rightarrow n. \text{ Velja tudi } \Psi_{n\vec{k}}(\vec{r} + \vec{R}) = e^{i\vec{k}\vec{R}} \Psi_{n\vec{k}}(\vec{r})$$

Dokaz: $T_{\vec{R}} \Psi(\vec{r}) = \Psi(\vec{r} + \vec{R})$ operator translacije

$$T_{\vec{R}} H \Psi = H(\vec{r} + \vec{R}) \Psi(\vec{r} + \vec{R}) = H(\vec{r}) \Psi(\vec{r} + \vec{R}) = H T_{\vec{R}} \Psi \Rightarrow [H, T_{\vec{R}}] = 0$$

⇒ Lastna stanja H morajo biti tudi lastna stanja $T_{\vec{R}}$: $H \Psi = \epsilon \Psi$

$$T_{\vec{R}} \Psi = c(\vec{R}) \Psi$$

$$T_{\vec{R}'} T_{\vec{R}} \Psi = c(\vec{R}) c(\vec{R}') \Psi, \quad \Rightarrow c(\vec{R} + \vec{R}') = c(\vec{R}) c(\vec{R}')$$

$$T_{\vec{R}} T_{\vec{R}'} \Psi = T_{\vec{R} + \vec{R}'} \Psi = c(\vec{R} + \vec{R}') \Psi$$

$\vec{a}_i$: primitivni vektorji Brav. reš. $C(\vec{a}_i) = e^{2\pi i x_i}$

$$\rightarrow C(\vec{r}) = C(\vec{a}_1)^{n_1} C(\vec{a}_2)^{n_2} C(\vec{a}_3)^{n_3} = e^{i\vec{k}\cdot\vec{r}}, \quad \vec{k} = x_1 \vec{A}_1 + x_2 \vec{A}_2 + x_3 \vec{A}_3$$

$$\Rightarrow T_{\vec{r}} \psi = C(\vec{r}) \psi = e^{i\vec{k}\cdot\vec{r}} \psi(\vec{r})$$

13. Elektronska stanja v približku šibkega periodičnega potenciala

~~$$U_{\vec{k}} = \frac{1}{V} \int_{\text{cel.}} e^{-i\vec{k}\cdot\vec{r}} U(\vec{r}) d\vec{r}, \quad U_0 = 0$$~~

$$U_{-\vec{k}} = U_{\vec{k}}^*. \quad \text{Če velja } U(\vec{r}) = U(-\vec{r}): \quad U_{\vec{k}} = U_{-\vec{k}}$$

$$-\frac{\hbar^2}{2m} \nabla^2 \psi = \sum_{\vec{q}} \frac{\hbar^2}{2m} q^2 e^{i\vec{q}\cdot\vec{r}}, \quad U\psi = \sum_{\vec{k}, \vec{q}} c_{\vec{q}} U_{\vec{k}} e^{i(\vec{q}+\vec{k})\cdot\vec{r}} = \sum_{\vec{k}, \vec{q}} U_{\vec{k}} c_{\vec{q}-\vec{k}} e^{i\vec{q}\cdot\vec{r}}$$

$$\rightarrow \sum_{\vec{q}} e^{i\vec{q}\cdot\vec{r}} \left[\left(\frac{\hbar^2 q^2}{2m} - \varepsilon \right) c_{\vec{q}} + \sum_{\vec{k}} U_{\vec{k}} c_{\vec{q}-\vec{k}} \right] = 0$$

$\hookrightarrow$ ravni valovi so ortogonalni $\Rightarrow \left(\frac{\hbar^2 q^2}{2m} - \varepsilon \right) c_{\vec{q}} + \sum_{\vec{k}} U_{\vec{k}} c_{\vec{q}-\vec{k}} = 0, \quad \vec{q} = \vec{k} - \vec{k}'$
($\vec{k}$ znotraj 1.BC)

$$\psi_{\vec{k}} = \sum_{\vec{k}'} c_{\vec{k}-\vec{k}'} e^{i(\vec{k}-\vec{k}')\cdot\vec{r}} = e^{i\vec{k}\cdot\vec{r}} \sum_{\vec{k}'} c_{\vec{k}-\vec{k}'} e^{-i\vec{k}'\cdot\vec{r}} = e^{i\vec{k}\cdot\vec{r}} U_{\vec{k}}(\vec{r})$$

$$\Rightarrow \left[\frac{\hbar^2}{2m} (\vec{k}-\vec{k}')^2 - \varepsilon \right] c_{\vec{k}-\vec{k}'} + \sum_{\vec{k}'} U_{\vec{k}-\vec{k}'} c_{\vec{k}-\vec{k}'} = 0$$

a) Prosti elektroni: $U_{\vec{k}} = 0 \rightarrow (\varepsilon_{\vec{k}-\vec{k}'}^0 - \varepsilon) c_{\vec{k}-\vec{k}'} = 0, \quad \varepsilon_{\vec{q}}^0 = \frac{\hbar^2 q^2}{2m}$

$$\Rightarrow \varepsilon = \varepsilon_{\vec{q}}^0 \quad \text{in} \quad \psi_{\vec{k}} \propto e^{i(\vec{k}-\vec{k}')\cdot\vec{r}}$$

b) $|U_{\vec{k}}| \ll \varepsilon_{\vec{k}-\vec{k}_1}^0$

Nedog. primer: izberemo $\vec{k}_1$: $|\varepsilon_{\vec{k}-\vec{k}_1}^0 - \varepsilon_{\vec{k}-\vec{k}}^0| \ll U_{\vec{k}-\vec{k}_1} \Rightarrow c_{\vec{k}-\vec{k}} \approx 0, \quad c_{\vec{k}-\vec{k}_1} \approx 1$
($\vec{k} \neq \vec{k}_1$)

$$(\varepsilon_{\vec{k}-\vec{k}}^0 - \varepsilon) c_{\vec{k}-\vec{k}} + c_{\vec{k}-\vec{k}_1} U_{\vec{k}-\vec{k}_1} = 0 \Rightarrow c_{\vec{k}-\vec{k}} = -c_{\vec{k}-\vec{k}_1} \frac{U_{\vec{k}-\vec{k}_1}}{\varepsilon_{\vec{k}-\vec{k}}^0 - \varepsilon}$$

$$(\varepsilon_{\vec{k}-\vec{k}_1}^0 - \varepsilon) c_{\vec{k}-\vec{k}_1} + \sum_{\vec{k}} c_{\vec{k}-\vec{k}} U_{\vec{k}-\vec{k}_1} = 0$$

$$\Rightarrow \varepsilon = \varepsilon_{\vec{k}-\vec{k}_1}^0 - \sum_{\vec{k}} \frac{|U_{\vec{k}-\vec{k}_1}|^2}{\varepsilon_{\vec{k}-\vec{k}}^0 - \varepsilon} \approx \varepsilon_{\vec{k}-\vec{k}_1}^0 - \sum_{\vec{k}} \frac{|U_{\vec{k}-\vec{k}_1}|^2}{\varepsilon_{\vec{k}-\vec{k}}^0 - \varepsilon_{\vec{k}-\vec{k}_1}^0}$$

Degeneriran primer: $|\varepsilon_{\vec{k}-\vec{k}_1}^0 - \varepsilon_{\vec{k}-\vec{k}_2}^0| \ll |U_{\vec{k}_1-\vec{k}_2}|$

$$(\varepsilon_{\vec{k}-\vec{k}_1}^0 - \varepsilon) C_{\vec{k}-\vec{k}_1} + U_{\vec{k}_2-\vec{k}_1} C_{\vec{k}-\vec{k}_2} = 0, (\varepsilon_{\vec{k}-\vec{k}_2}^0 - \varepsilon) C_{\vec{k}-\vec{k}_2} + U_{\vec{k}_1-\vec{k}_2} C_{\vec{k}-\vec{k}_1} = 0$$

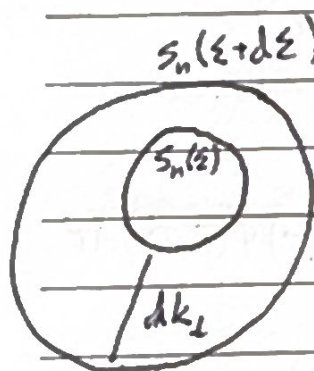
$$\vec{q} = \vec{k} - \vec{k}_1, \vec{k} = \vec{k}_2 - \vec{k}_1 \rightarrow \begin{vmatrix} \varepsilon - \varepsilon_q^0 & -U_{\vec{k}} \\ U_{\vec{k}}^* & \varepsilon - \varepsilon_{q-\vec{k}}^0 \end{vmatrix} = 0$$

$$\varepsilon_q^{\pm} = \frac{1}{2}(\varepsilon_q^0 + \varepsilon_{q-\vec{k}}^0) \pm \frac{1}{2} \sqrt{(\varepsilon_q^0 - \varepsilon_{q-\vec{k}}^0)^2 + 4|U_{\vec{k}}|^2} \approx \varepsilon_q^0 \pm |U_{\vec{k}}|$$

14. Gostota el. stanj

$$g = \frac{2^d}{V} \sum_n Q(\varepsilon_n) = \sum_n \int \frac{d^3 k}{4\pi^3} Q(\varepsilon_n(\vec{k})) = \int d\varepsilon g(\varepsilon) Q(\varepsilon), \quad g(\varepsilon) = \sum g_n(\varepsilon)$$

$$\rightarrow g(\varepsilon) = \int \delta(\varepsilon - \varepsilon_n(\vec{k})) \frac{d^3 k}{4\pi^3} \rightarrow g_n(\varepsilon) d\varepsilon = \begin{cases} \int \frac{d^3 k}{4\pi^3}, & \varepsilon < \varepsilon_n(\vec{k}) < \varepsilon + d\varepsilon \\ 0, & \text{sicer} \end{cases}$$



$$d\varepsilon = \nabla \varepsilon_n(\vec{k}) \cdot d\vec{k} = |\nabla \varepsilon_n(\vec{k})| dk_{\perp}$$

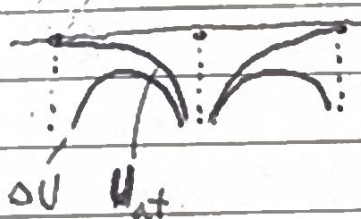
$$g(\varepsilon) d\varepsilon = \int \frac{d^3 k}{4\pi^3} = \frac{1}{4\pi^3} \int dS dk_{\perp} = \frac{1}{4\pi^3} \int \frac{dS d\varepsilon}{|\nabla \varepsilon_n(\vec{k})|}$$

$$\Rightarrow g(\varepsilon) = \frac{1}{4\pi^3} \int \frac{dS}{|\nabla \varepsilon_n(\vec{k})|}$$

15. Metoda tesne vezi

- predp.: atomske val. funke. se šibko prekrivajo med sabo

$$H_{at} \Psi_n = E_n \Psi_n \quad n\text{-ta atomska funkcija} \quad H = H_{at} + \Delta U(\vec{r})$$



$$\Psi_{n\vec{k}}(\vec{r}) = \sum_{\vec{R}} e^{i\vec{k}\vec{R}} \Psi_n(\vec{r}-\vec{R}), \quad \Psi_{n\vec{k}}(\vec{r}+\vec{R}) = e^{i\vec{k}\vec{R}} \Psi_{n\vec{k}}(\vec{r})$$

Nastavek: $\Psi(\vec{r}) = \sum_n b_n \Psi_n(\vec{r})$

$\rightarrow \Psi(\vec{r}) = \sum_{\vec{R}} e^{i\vec{k}\vec{R}} \Psi(\vec{r}-\vec{R}) = \sum_{n, \vec{R}} b_n e^{i\vec{k}\vec{R}} \Psi_n(\vec{r}-\vec{R})$

$(H_{at} + \Delta U) \Psi = \varepsilon(t) \Psi(\vec{r}) \quad / \int \Psi_m^* d^3\vec{r}$

$\cdot \int \underbrace{\Psi_m^*(\vec{r})}_{E_m \Psi_m} H_{at} \Psi(\vec{r}) d^3\vec{r} = \cancel{\int \Psi_m^* \Psi d^3\vec{r}} E_m \int \Psi_m^* \Psi(\vec{r}) d^3\vec{r}$

$\cdot \int \Psi_m^* \Psi(\vec{r}) d^3\vec{r} = \sum_n b_n \int \Psi_m^* \Psi_n d^3\vec{r} + \sum_{n, \vec{R} \neq 0} b_n e^{i\vec{k}\vec{R}} \underbrace{\int \Psi_m^*(\vec{r}) \Psi_n(\vec{r}-\vec{R}) d^3\vec{r}}_{\alpha_{mn}(\vec{R})}$

$\cdot \int \Psi_m^* \Delta U \Psi d^3\vec{r} = \sum_n b_n \underbrace{\int \Psi_m^* \Delta U \Psi_n d^3\vec{r}}_{-\beta_{mn}(\vec{r})} + \sum_{n, \vec{R} \neq 0} b_n e^{i\vec{k}\vec{R}} \underbrace{\int \Psi_m^*(\vec{r}) \Delta U(\vec{r}) \Psi_n(\vec{r}-\vec{R}) d^3\vec{r}}_{-\gamma_{mn}(\vec{R})}$

Poenostavitev: upoštevamo samo prekrivanje s-orbital

$\Psi \Psi_0 = b_0 \Psi_0, E_n = E_0 \delta_{n0} \rightarrow (\varepsilon(t) - E_0) \left(1 + \sum_{\vec{R} \neq 0} \alpha(\vec{R}) e^{i\vec{k}\vec{R}}\right) = -\beta - \sum_{\vec{R} \neq 0} \gamma(\vec{R}) e^{i\vec{k}\vec{R}}$

$\Rightarrow \varepsilon(t) = E_0 - \frac{\beta + \sum e^{i\vec{k}\vec{R}} \gamma(\vec{R})}{1 + \sum e^{i\vec{k}\vec{R}} \alpha(\vec{R})}$

$\alpha(\vec{R}) = \int \Psi^*(\vec{r}) \Psi(\vec{r}-\vec{R}) d^3\vec{r}, \beta = -\int \Delta U(\vec{r}) |\Psi(\vec{r})|^2 d^3\vec{r}, \gamma(\vec{R}) = -\int \Psi^*(\vec{r}) \Delta U(\vec{r}) \Psi(\vec{r}-\vec{R}) d^3\vec{r}$

16. Semiklasični opis gibanja e^- v periodičnem potencialu, efektivna masa,
 $\vec{r} = \frac{\hbar \vec{k}}{m}, \hbar \dot{\vec{k}} = \vec{F} = -e(\vec{E} + \vec{v} \times \vec{B})$ opisuje gibanje val. paketa $\Psi(\vec{r}, t) = \sum_{\vec{k}'} g(\vec{k}') e^{i(\vec{k}'\vec{r} - \frac{\hbar k'^2}{2m}t)}$

Velja $g(\vec{k}') \approx 0$ za $|\vec{k}' - \vec{k}| > \Delta k \Rightarrow \Delta k \ll \text{dimenzija 1.BC} \Rightarrow \text{val. paket} \gg a, \Delta R \sim \frac{1}{\Delta k}$

Predpostavke: - n se ohranja

$\vec{r} = \vec{v}_n(\vec{k}) = \frac{1}{\hbar} \frac{\partial \varepsilon_n}{\partial \vec{k}}, \hbar \dot{\vec{k}} = -e[\vec{E}(\vec{r}, t) + \vec{v}_n(\vec{k}) \times \vec{B}(\vec{r}, t)]$

grupna hitrost - $\vec{k}$ določen le do poljubnega $\vec{k}$

$\cdot \vec{B} = 0: E = -\nabla \phi \Rightarrow \varepsilon_n(\vec{k}) - e\phi = \text{konst.} / \frac{d}{dt}$

$\frac{\partial \varepsilon_n}{\partial \vec{k}} \vec{k} - e \nabla \phi \vec{r} = 0 = \hbar \vec{v}_n \dot{\vec{k}} - e \vec{v}_n \nabla \phi \Rightarrow \hbar \dot{\vec{k}} = -e \vec{E}$

Efektivna masa: razvoj okoli minimuma(e)/maksimuma(h): $\varepsilon = \varepsilon_0 \pm A(\vec{k} - \vec{k}_0)^2$
 $A = \frac{\hbar^2}{2m^*}$
 $\hookrightarrow \text{ef. masa}$

splošno: $\varepsilon(\vec{k}_0 + \delta \vec{k}) = \varepsilon(\vec{k}_0) + \sum \frac{\partial \varepsilon}{\partial k_i} \delta k_i + \frac{1}{2} \sum_{ij} \frac{\partial^2 \varepsilon}{\partial k_i \partial k_j} \delta k_i \delta k_j + \dots$

$\rightarrow M_{ij}^{-1}(\vec{k}) = \frac{\hbar^2}{e} \frac{\partial^2 \varepsilon(\vec{k})}{\partial k_i \partial k_j}$ (o(razvoj ok. min.))
 Tenzor efektivne mase

17. Blochove oscilacije, gibanje e^- v $\vec{B} \neq 0$

$$\hbar \dot{\vec{k}} = -e\vec{E}, \quad \vec{v}(t) = \frac{1}{\hbar} \frac{\partial \varepsilon(\vec{k})}{\partial \vec{k}}, \quad \vec{E}(\vec{r}, t) = \vec{E} \text{ (konst. el. polje)}$$

$$\rightarrow \vec{k}(t) = \vec{k}(0) - \frac{e\vec{E}t}{\hbar} \text{ narašča linearno.}$$

$$1D, \quad \varepsilon(k) = -\gamma \cos ka \rightarrow v_k = \frac{\gamma a}{\hbar} \sin ka$$

$$\text{Blochova perioda: } \frac{eE t_B}{\hbar} = 2\pi \rightarrow \underline{t_B = \frac{h}{eEa}}$$

Omejitve semiklasичnega modela

- ne predvideva prehodov med el. pasovi
- popolnjen pas ne prispeva k el. toku

$$\vec{B} = B\hat{e}_z \quad k_z \text{ je konstanta gibanja: } \dot{k}_z = 0, \text{ ker } \hat{e}_z \perp \vec{v} \times \vec{B}$$

$$\dot{\varepsilon}(\vec{k}) = \vec{v} \cdot \hbar \dot{\vec{k}} = \vec{v} \cdot (\vec{v} \times \vec{B}) = 0 \Rightarrow \varepsilon(\vec{k}) \text{ konst. gib.}$$

$$\vec{v} = \frac{1}{\hbar} \frac{\partial \varepsilon(\vec{k})}{\partial \vec{k}} = \frac{1}{\hbar} \frac{\hbar^2}{2} 2m^* \vec{k} = \hbar m^* \vec{k} \rightarrow m^* \vec{v} = \hbar \vec{k}, \quad \vec{v} = \vec{v}_0 e^{i\omega t}$$

$$-i\omega m^* \vec{v}_0 = e \vec{v}_0 \times \vec{B}$$

$$\begin{pmatrix} m_{xx} & m_{xy} & m_{xz} \\ m_{xy} & m_{yy} & m_{yz} \\ m_{xz} & m_{yz} & m_{zz} \end{pmatrix} \begin{pmatrix} v_{0x} \\ v_{0y} \\ v_{0z} \end{pmatrix} = i \frac{e}{\omega} \begin{pmatrix} v_{0y} B \\ v_{0x} B \\ 0 \end{pmatrix} \rightarrow \det(\dots) = 0 = \det(m^*) - m_{zz} \left(\frac{e}{\omega} B \right)^2$$

$$\rightarrow \omega = |e|B \sqrt{\frac{m_{zz}}{\det(m^*)}}$$

18., 19. Nismo delali

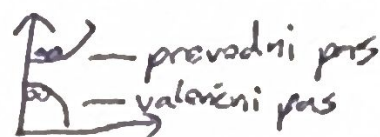
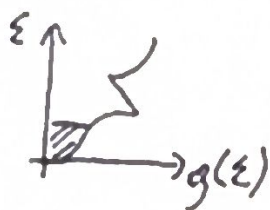
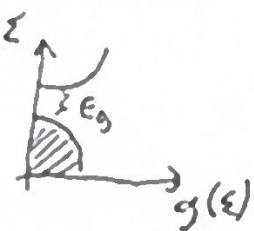
29. Homogeni polprevodniki, razlika med prevodniki, polprevodniki, izolatorji

zapolnjen $\leftrightarrow$ delno zapolnjen pas

izolator

kovina

E_g ... energijska reža



Polprevodnik je izolator le pri $T=0$, končna T :

Število e^- v prev. pasu: $n_c \propto e^{-E_g/2k_B T}$

Spec. upornost:

	Kovina	polprev.	izolator
$\rho [\Omega \cdot m]$	10^{-6}	$10^{-3} - 10^3$	10^{22}

Kovina

Polprevodnik

$\sigma = \frac{ne^2\tau}{m}$ se manjša

σ močno narašča z večanjem T , ker $n_c \propto \exp(-E_g/2k_B T)$

z naraščanjem T

τ pada z nar. T

21. Gostota e^- v prevodnem (valenčnem) pasu homogenega ter čistega pdpr.

$$\epsilon_c(\mathbf{k}) = \epsilon_c \pm \frac{1}{2} \hbar^2 \sum_{ij} \delta k_i (M^{-1})_{ij} \delta k_j \quad M^{-1} \text{ realna, simetrična} \Rightarrow$$

$$\epsilon_c(\mathbf{k}) = \epsilon_c(\mathbf{k}_0) \pm \frac{\hbar^2}{2} \left(\frac{k_1^2}{m_1} + \frac{k_2^2}{m_2} + \frac{k_3^2}{m_3} \right)$$

$$n_c(T) = \int_{\epsilon_c}^{\infty} g_c(\epsilon) \frac{1}{e^{(\epsilon - \mu)/k_B T} + 1} d\epsilon, \quad p_v(T) = \int_{-\infty}^{\epsilon_v} g_v(\epsilon) \frac{1}{e^{(\mu - \epsilon)/k_B T} + 1} d\epsilon, \quad \text{predp. } \epsilon_c - \mu \gg k_B T, \mu - \epsilon_v \gg k_B T$$

$$\Rightarrow n_c(T) = \int_{\epsilon_c}^{\infty} g_c(\epsilon) e^{-(\epsilon - \mu)/k_B T} d\epsilon = \int_{\epsilon_c}^{\infty} g_c(\epsilon) \exp(-(\epsilon - \epsilon_c)/k_B T) d\epsilon \cdot \exp(-(\epsilon_c - \mu)/k_B T) = \underline{N_c(T) e^{-\frac{(\epsilon_c - \mu)}{k_B T}}}$$

$$p_v(T) = \dots = P_v(T) e^{-\frac{\mu - \epsilon_v}{k_B T}}$$

$$g_c(\epsilon) = \sqrt{2|\epsilon - \epsilon_c|} \frac{m_c^{3/2}}{\pi^2 \hbar^3}$$

$$N_c(T) = \dots = \frac{\sqrt{2}}{\pi^2} \left(\frac{m_c k_B T}{\hbar^2} \right)^{3/2} \underbrace{\int_0^{\infty} x^{1/2} e^{-x} dx}_{\frac{\sqrt{\pi}}{2}} = \frac{1}{4} \left(\frac{2m_c k_B T}{\pi \hbar^2} \right)^{3/2}$$

$$P_v(T) = \frac{1}{4} \left(\frac{2m_v k_B T}{\pi \hbar^2} \right)^{3/2}$$

$$n_c p_v = N_c P_v e^{-E_g/k_B T} \neq f(\mu)$$

Čist polprevodnik: $n_c = p_v = n_i$

$$n_i(T) = \sqrt{N_c p_v} e^{-E_g/2k_B T} = \frac{1}{4} \left(\frac{2k_B T}{\pi \hbar^2} \right)^{3/4} (m_c m_v)^{1/4} e^{-E_g/2k_B T}$$

22. Temp. odv. kemijskega potenciala

$$n_c = p_v \Rightarrow N_c e^{-(\epsilon_c - \mu)/k_B T} = p_v e^{-(\mu - \epsilon_v)/k_B T}, \quad \epsilon_c = \epsilon_v + E_g$$

$$\rightarrow \mu = \frac{\epsilon_c + \epsilon_v}{2} + \frac{1}{2} k_B T \ln \frac{p_v}{N_c} = \epsilon_v + \frac{1}{2} E_g + \frac{3}{4} k_B T \ln \frac{m_v}{m_c}$$

23. Opis energijskih stanj nečistoč ter njihov vpliv na gostoti nosilcev v prevodnem in valenčnem pasu

$$n_c - p_v = \Delta n, \quad n_c p_v = n_i^2 \Rightarrow \left\{ \begin{matrix} n_c \\ p_v \end{matrix} \right\} = \frac{1}{2} \sqrt{\Delta n^2 + 4n_i^2} \pm \frac{1}{2} \Delta n$$

$\frac{\Delta n}{n_i}$... merilo za vpliv nečistoč

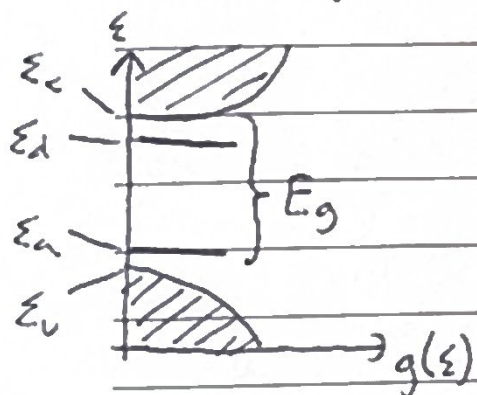
$$n_c = e^{\beta(\mu - \mu_i)}, \quad p_v = e^{-\beta(\mu - \mu_i)} \Rightarrow \frac{\Delta n}{n_i} = 2 \sinh \beta(\mu - \mu_i), \quad \mu_i \dots \text{kem. pot. čistega polpr.}$$

Nivoji nečistoč

• donorji: prispevajo dod. e^- v prev. pas

• akceptorji: prispevajo dod. h^+ v valenčni pas

Bohrov radij nečistoče: $r_0 = \frac{m}{m^*} \epsilon a_0$, $\epsilon_b = \frac{m^*}{m} \frac{1}{\epsilon^2} R_y$, $R_y = 13,6 \text{ eV}$; $a_0 = 0,53 \text{ Å}$
 $\hookrightarrow$ diel. konst.



$$\epsilon_c - \epsilon_d \ll E_g, \quad \epsilon_a - \epsilon_v \ll E_g$$

24. Zasedenost donorskih / akceptorskih nivojev ter ravnovesne gostote nosilcev nabojev

Donorski nivo: 3 stanja: $|0\rangle, |\uparrow\rangle, |\downarrow\rangle$

$$\langle n \rangle = \frac{2 e^{-\beta(\epsilon_d - \mu)}}{1 + 2 e^{-\beta(\epsilon_d - \mu)}} = \frac{1}{\frac{1}{2} e^{\beta(\epsilon_d - \mu)} + 1} \rightarrow n_d = N_d \langle n \rangle = \frac{N_d}{\frac{1}{2} e^{\beta(\epsilon_d - \mu)} + 1}$$

Akceptorski nivo:

$$\langle n \rangle = \frac{2 e^{\beta\mu} + 2 e^{-\beta(\epsilon_a - \mu)}}{2 e^{\beta\mu} + e^{-\beta(\epsilon_a - \mu)}} = \frac{e^{\beta(\mu - \epsilon_a)} + 1}{\frac{1}{2} e^{\beta(\mu - \epsilon_a)} + 1} \quad \langle p \rangle = 2 - \langle n \rangle = \frac{1}{\frac{1}{2} e^{\beta(\mu - \epsilon_a)} + 1}$$

$$p_a = N_a \langle p \rangle = \frac{N_a}{\frac{1}{2} e^{\beta(\mu - \epsilon_a)} + 1}$$

$N_d > N_a$:

$T = 0$:

$$n_d = N_d - N_a, \quad n_c = 0$$

$$p_a = p_v = 0$$

$T \neq 0$:

ohranitev naboja: $n_d + n_c = N_d - N_a + p_v + p_a$

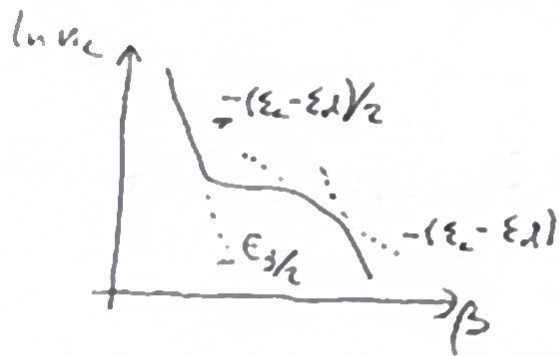
Približek: $\epsilon_d - \mu \gg k_B T, \mu - \epsilon_a \gg k_B T \Rightarrow n_d \ll N_d, p_a \ll N_a$

$$n_c - p_v = N_d - N_a, \quad n_c p_v = n_i^2$$

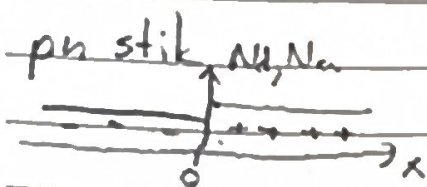
$$\rightarrow \begin{cases} n_c \\ p_v \end{cases} = \frac{1}{2} \sqrt{(N_d - N_a)^2 + 4 n_i^2} \pm \frac{1}{2} (N_d - N_a), \quad \frac{N_d - N_a}{n_i} = 2 \sinh(\beta(\mu - \mu_i))$$

• limita čistega polpr. ($n_i \gg N_d - N_a$): $\begin{cases} n_c \\ p_v \end{cases} = n_i \pm \frac{1}{2} (N_d - N_a)$

• limita nočno doprane polpr.: $n_c = N_d - N_a, \quad p_v = n_i^2 / (N_d - N_a)$



25. Nehomogeni polpr., zaporna plast ter vpliv el. pot. na gostoto nosilcev v prevodnem in valenčnem pasu



$$N_d(x) = \begin{cases} N_D; & x > 0 \\ 0; & x < 0 \end{cases} \quad N_A(x) = \begin{cases} 0; & x > 0 \\ N_A; & x < 0 \end{cases}$$

$\Rightarrow n_c = n_c(x), p_v = p_v(x), \phi = \phi(x)$. Krajevna odvisnost n_c, p_v, ϕ je omejena na okolico $x=0 \Rightarrow$ zaporna plast $\sim 10^{-10^3}$ nm

Semiklasični model: $H_n = \xi_n - e\phi(x)$. Pogoj: $e\Delta\phi \ll E_g$
↳ indeks pasu

TD ravnovesje $\Rightarrow \mu$ enak na obeh straneh, $\xi_c \rightarrow \xi_c - e\phi; \xi_v \rightarrow \xi_v - e\phi$

$$n_c(x) = N_c(T) e^{-\beta(\xi_c - e\phi(x) - \mu)}, \quad p_v(x) = P_v(T) e^{-\beta(\mu - \xi_v + e\phi(x))}$$

26. Odvisnost skoka el. pot. prek p-n stika od gostote nečistoč. Potek pasov v bližini stika

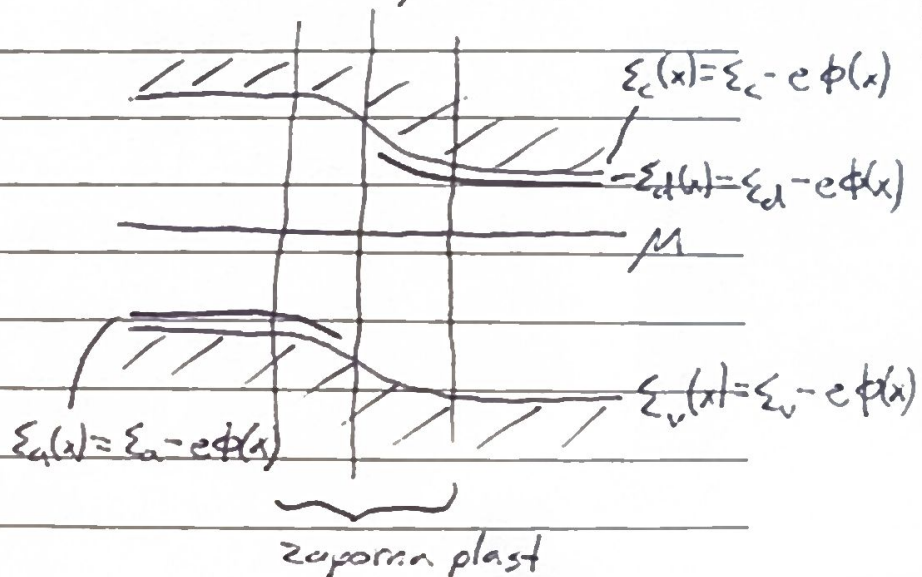
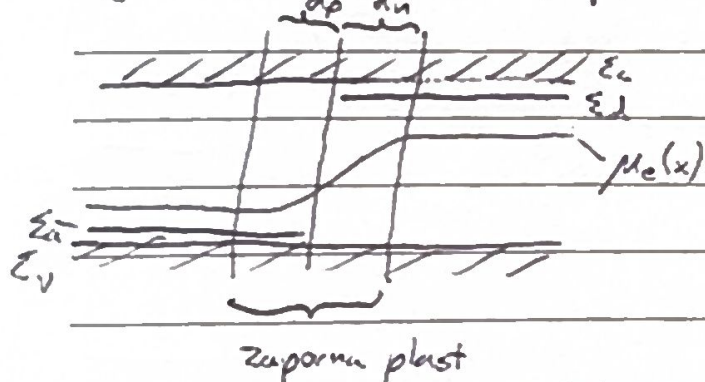
Predp.: daleč od stika so nečistoče popolnoma ionizirane:

$$N_d = n_c(\infty) = N_c(T) e^{-\beta(\xi_c - e\phi(\infty) - \mu)}, \quad N_A = p_v(-\infty) = P_v(T) e^{-\beta(\mu - \xi_v + e\phi(-\infty))}$$

$$\rightarrow e(\phi(\infty) - \phi(-\infty)) = e\Delta\phi = \xi_c - \xi_v + k_B T \ln \frac{N_d N_A}{N_c P_v} = E_g + k_B T \ln \frac{N_d N_A}{N_c P_v}$$

≡ elektro-kenijski potencial: $\mu_c(x) = \mu + e\phi(x)$

$$n_c(x) = N_c(T) e^{-\beta(\xi_c - \mu_c(x))}, \quad p_v(x) = P_v(T) e^{-\beta(\mu_c(x) - \xi_v)}, \quad e\Delta\phi = \mu_c(\infty) - \mu_c(-\infty)$$



27. Obliku potenciala prek p-n stika v limiti pop. ioniziranih nečistož

Poisson: $-\nabla^2 \phi = \frac{\rho(x)}{\epsilon \epsilon_0}$, $\rho(x) = e[N_d(x) - N_a(x) - n_c(x) + p_v(x)]$

$$n_c(x) = N_d e^{-\beta e(\phi(x) - \phi(x))}$$

$$p_v(x) = N_a e^{-\beta e(\phi(x) - \phi(-\infty))}$$

$$\rho(x) = \begin{cases} 0; & x > d_n \\ eN_d; & 0 < x < d_n \\ -eN_a; & -d_p < x < 0 \\ 0; & x < -d_p \end{cases}$$

$$\phi''(x) = \begin{cases} 0; & x > d_n \\ -\frac{eN_d}{\epsilon \epsilon_0}; & 0 < x < d_n \\ \frac{eN_a}{\epsilon \epsilon_0}; & -d_p < x < 0 \\ 0; & x < -d_p \end{cases}$$

$$\Rightarrow \phi(x) = \begin{cases} \phi(\infty); & x > d_n \\ \phi(\infty) - \frac{eN_d}{2\epsilon \epsilon_0} (x - d_n)^2; & 0 < x < d_n \\ \phi(-\infty) + \frac{eN_a}{2\epsilon \epsilon_0} (x + d_p)^2; & -d_p < x < 0 \\ \phi(-\infty); & x < -d_p \end{cases}$$

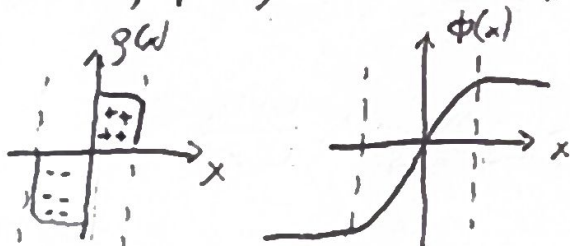
zvezna odvedljivost pri $x=0$: $N_d d_n = N_a d_p$

zveznost pri $x=0$: $\frac{e}{2\epsilon \epsilon_0} (N_d d_n^2 + N_a d_p^2) = \phi(\infty) - \phi(-\infty) = \Delta \phi$

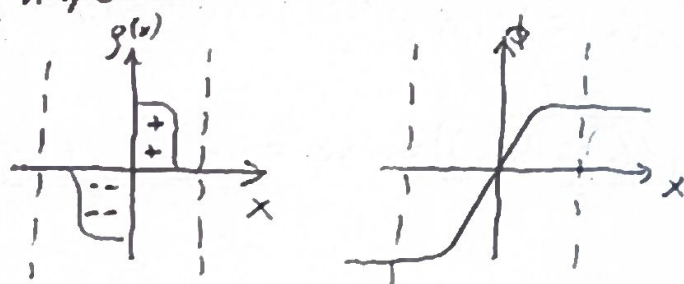
$$\Rightarrow d_{n,p} = \sqrt{\frac{2\epsilon \epsilon_0 \Delta \phi}{e}} \sqrt{\left(\frac{N_a}{N_d}\right)^{\pm 1} \frac{1}{N_d \mp N_a}}$$

28. P-N stik, priključen na napetost, vpliv napetosti na širino zaporne plasti

$V=0$:



$V>0$:



$$d_{n,p}(V) = d_{n,p}(V=0) \sqrt{1 - \frac{V}{\Delta \phi_0}} \quad \Delta \phi = \Delta \phi_0 - V$$

29. IV karakteristika p-n stika

$$\Delta\phi = \Delta\phi_0 - V, \quad d_{n,p}(V) = d_{n,p}(V=0) \sqrt{1 - V/\Delta\phi_0}$$

$$j_e = -e J_e, \quad j_h = e J_h$$

↳ el. tok ↳ št. tok

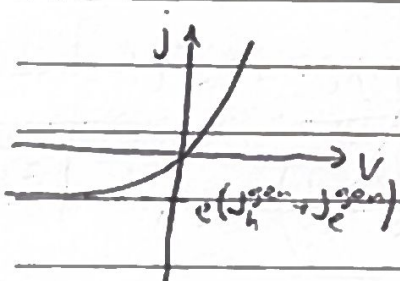
$$V=0: \quad \langle J_e \rangle = \langle J_h \rangle = 0$$

$V \neq 0$: Generacijski tok: vsako vrzel na n-strani ($p_v \sim \frac{n_i}{N_A}$), ki nastane zaradi termične ekscitacije, $\Delta\phi$ posrka na p-stran podobno e^- na p-strani

Rekombinacijski tok: vrzeli lahko prehajajo v smeri $p \rightarrow n$, če imajo dovolj termične energije: $J_h^{\text{rec}} \propto e^{-\beta e(\Delta\phi_0 - V)}$

veljati mora: $J_h^{\text{rec}}|_{V=0} = J_h^{\text{gen}} \Rightarrow J_h^{\text{rec}} = J_h^{\text{gen}} e^{\beta e V}$

$$\rightarrow J_h = J_h^{\text{rec}} - J_h^{\text{gen}} = J_h^{\text{gen}} (e^{\beta e V} - 1) \rightarrow j = e (J_h^{\text{gen}} + J_e^{\text{gen}}) (e^{\beta e V} - 1)$$



30. 1D model za opis mrežnih nihanj, disperzijska relacija, val. en.

$$U^{\text{larm}} = \frac{K}{2} \sum_n (u_n - u_{n+1})^2 \quad \text{sila na n-ti atom: } F_n = -\frac{\partial U^{\text{larm}}}{\partial u_n} = -K(2u_n - u_{n-1} - u_{n+1})$$

$$\rightarrow M \ddot{u}_n = F_n = K(u_{n+1} + u_{n-1} - 2u_n) \quad \text{Nastavek: } u_n = u_0 e^{ikna} e^{i\omega t}$$

$$-\omega^2 M u_0 e^{ikna} = K(u_0 e^{ik(n+1)a} + u_0 e^{ik(n-1)a} - 2u_0 e^{ikna})$$

$$\rightarrow \omega^2 = \frac{K}{M} 2(1 - \cos ka) = \frac{K}{M} 4 \sin^2 \frac{ka}{2} \rightarrow \omega = 2 \sqrt{\frac{K}{M}} \left| \sin \frac{ka}{2} \right|$$

Disperzijska relacija: zveza med ω in k

$$u_n \rightarrow u(x), \quad x=na, \quad a \rightarrow 0$$

$$M\ddot{u}_n = K(u_{n+1} + u_{n-1} - 2u_n) \rightarrow M \frac{\partial^2 u}{\partial t^2} = Ka^2 \lim_{a \rightarrow 0} \frac{\frac{u(x+a)-u(x)}{a} - \frac{u(x)-u(x-a)}{a}}{a} = Ka^2 \frac{\partial^2 u}{\partial x^2}$$

$$\rightarrow \frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \text{ valovna enačba, } c = a \sqrt{\frac{K}{M}}$$

31. Dvatomni 1D model za opis mrežnih nihanj, optična in akustična veja

$$\begin{array}{ccccccc} M & m & K & M & m \\ \bullet & \bullet & \bullet & \bullet & \bullet \\ u_n & v_n & u_{n+1} & v_{n+1} \end{array}$$

$$U^{\text{harmon}} = \frac{K}{2} \sum (u_n - v_n)^2 + (v_n - u_{n+1})^2$$

$$F_{u_n} = -K(u_n - v_n - (v_{n-1} - u_n))$$

$$F_{v_n} = -K(v_n - u_n + v_n - u_{n+1})$$

$$M\ddot{u}_n = -K(2u_n - v_n - v_{n+1}) \quad m\ddot{v}_n = -K(2v_n - u_n - u_{n+1})$$

$$\text{Nastavek: } u_n = u_0 e^{ikna} e^{-i\omega t}, \quad v_n = v_0 e^{ikna} e^{-i\omega t}$$

$$M\omega^2 u_0 = K(2u_0 - v_0 - v_0 e^{-ika}), \quad m\omega^2 v_0 = K(2v_0 - u_0 - u_0 e^{ika})$$

$$\rightarrow \begin{bmatrix} M\omega^2 - 2K & K(1 + e^{-ika}) \\ K(1 + e^{ika}) & m\omega^2 - 2K \end{bmatrix} \begin{bmatrix} u_0 \\ v_0 \end{bmatrix} = 0 \Rightarrow \det(\dots) = 0$$

$$(M\omega^2 - 2K)(m\omega^2 - 2K) - 2K^2(1 + \cos ka) = 0 = mM\omega^4 - 2K(m+M)\omega^2 + 2K^2(1 - \cos ka)$$

$$\rightarrow \omega^2 = \frac{1}{2mM} \left(2K(m+M) \pm \sqrt{4K^2(m+M)^2 - 8K^2mM(1 - \cos ka)} \right) = K \left(\frac{1}{m} + \frac{1}{M} \right) \pm K \sqrt{\left(\frac{1}{m} + \frac{1}{M} \right)^2 - \frac{4}{mM} \sin^2 \frac{ka}{2}}$$

32. Oblika lastnih nihanj v $k=0, k=\pi/a$

Dolgovalovna limita: $ka \rightarrow 0$

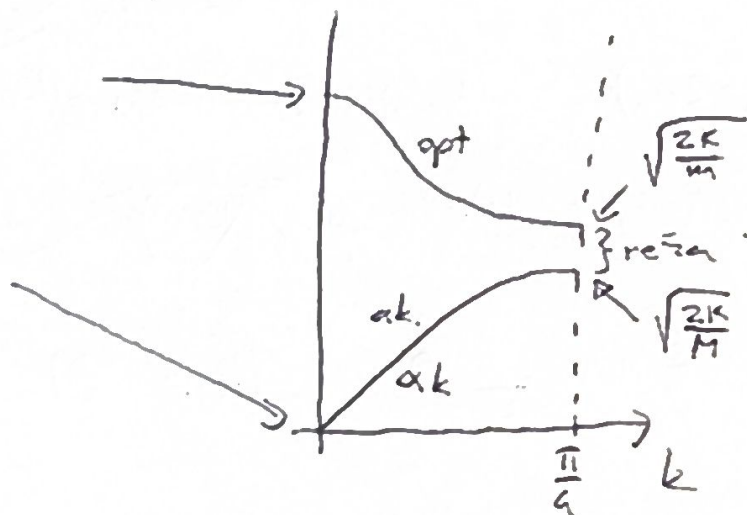
$$\omega^2 = k \left(\frac{1}{m} + \frac{1}{M} \right) \left[1 \pm \left(1 - \frac{k^2 a^2}{\left(\frac{1}{m} + \frac{1}{M} \right)^2 mM} \right)^{1/2} \right]$$

Optična veja (+):

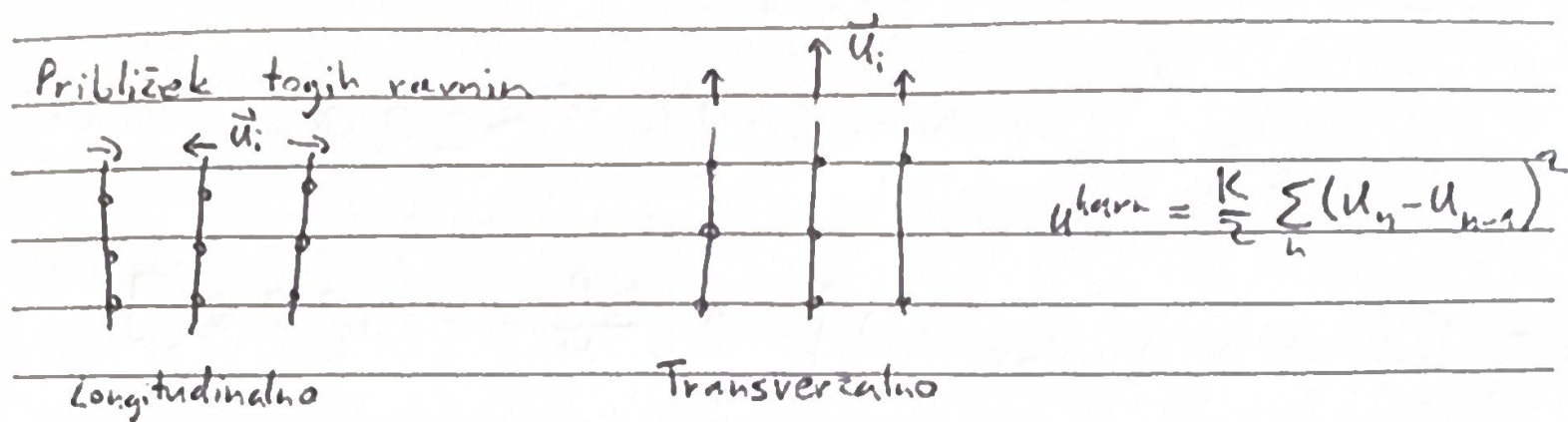
$$\omega^2 \approx 2k \left(\frac{1}{m} + \frac{1}{M} \right)$$

Akustična veja (-):

$$\omega^2 \approx \frac{\frac{1}{2}K}{m+M} k^2 a^2$$



33. Disperzijske relacije - opis nih. nač. v 3D v kristalih brez/baze



Relaksacije modela togih ravnin:

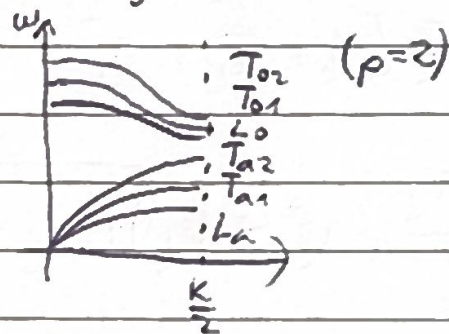
1) brez baze

- 1 atom na osn. celico, N atomov, $3D \rightarrow 3N$ prost. st., $3N$ last. nihanj, ampak le N različnih valovnih vektorjev $k \Rightarrow 3$ veje



2) 2 baza

- p atomov na osn. celico, $3pN$ prost. stopenj, $3(p-1)$ optičnih vej, 3 akustične veje
- Degeneracija T-vej je odvisna od smeri k in simetrije B.R.



34. Kvantizacija mrežnih nihanj, Hamiltonian za mrežna nihanja z bozonskimi anihilacijskimi in kreacijskimi operatorji

$$\hat{H} = \sum_n \frac{p_n^2}{2m} + \frac{K}{2} (u_n - u_{n+1})^2 \quad [u_n, p_{n'}] = i\hbar \delta_{nn'}$$

$$u_n = \frac{1}{\sqrt{N}} \sum_k Q_k e^{ikna}, \quad Q_k = \frac{1}{\sqrt{N}} \sum_n u_n e^{-ikna}$$

Periodični RP: $u_{N+1} = u_1$ in $k = \frac{2\pi}{Na} j$, $j = -(\frac{N}{2}-1), -(\frac{N}{2}-2), \dots, (\frac{N}{2}-1), \frac{N}{2}$

$$p_n = \frac{1}{\sqrt{N}} \sum_k P_k e^{ikna}, \quad P_k = \frac{1}{\sqrt{N}} \sum_n p_n e^{-ikna}$$

$$[Q_k, P_{k'}] = \frac{1}{N} \sum_{n,m} [u_n, p_m] e^{-i(kn - k'm)} a = i\hbar \frac{1}{N} \sum_n e^{-i(k-k')na} = i\hbar \delta_{k,k'}$$

$$\sum_n p_n^2 = \frac{1}{N} \sum_{n,k,k'} P_k P_{k'} \underbrace{e^{-i(k+k')na}}_{\delta_{k,-k'}} = \sum_k P_k P_{-k}$$

$$\sum_n (u_n - u_{n+a})^2 = \frac{1}{N} \sum_{n,k,k'} Q_k Q_{k'} e^{i(k+k')na} (1 - e^{ika}) (1 - e^{ik'a}) = \sum_k Q_k Q_{-k} 2(1 - \cos ka)$$

$$\rightarrow H = \sum_k \left[\frac{P_k P_{-k}}{2M} + k Q_k Q_{-k} (1 - \cos ka) \right] = \sum_k \left[\frac{P_k P_{-k}}{2M} + \frac{1}{2} M \omega_k^2 Q_k Q_{-k} \right]$$

$$i\hbar \dot{Q}_k = [Q_k, H] = \frac{1}{2M} [Q_k, \sum_{k'} P_{k'} P_{-k'}] = \frac{i\hbar}{M} P_{-k} \Rightarrow \dot{Q}_k = \frac{1}{M} P_{-k}$$

$$i\hbar \ddot{Q}_k = [\dot{Q}_k, H] = \frac{1}{M} [P_{-k}, \sum_{k'} \frac{M}{2} \omega_{k'}^2 Q_{k'} Q_{-k'}] = -i\hbar \omega_k^2 Q_k \Rightarrow$$

$$\Rightarrow \ddot{Q}_k + \omega_k^2 Q_k = 0 \text{ Enačba harmon. osc.}$$

Bozonski kreacijski in anihilacijski operator:

$$Q_k = \sqrt{\frac{\hbar}{2M\omega_k}} (a_k + a_{-k}^+), \quad P_k = i \sqrt{\frac{\hbar M \omega_k}{2}} (a_k^+ - a_{-k})$$

$$[Q_k, P_{k'}] = i \frac{\hbar}{2} [a_k + a_{-k}^+, a_{k'}^+ - a_{-k'}] = i\hbar \delta_{kk'} \checkmark$$

$$H = \sum_k -\frac{\hbar M \omega_k}{2 \cdot 2M} (a_k^+ - a_{-k})(a_{-k}^+ - a_k) + \frac{M \omega_k^2}{2} \frac{\hbar}{2M \omega_k} (a_k + a_{-k}^+)(a_{-k} + a_k^+) =$$

$$= \dots = \sum_k \hbar \omega_k (a_k^+ a_k + 1/2) = \sum_k \hbar \omega_k (n_k + 1/2) \checkmark$$

35. Fononski prispevek k specifični toploti (Nizko- in visko-T limita)

$$f = \frac{1}{V} \ln \sum_i e^{-\beta E_i}, \quad E_i = \sum_{\mathbf{k}, s} \hbar \omega_s(\mathbf{k}) (n_{\mathbf{k}, s} + \frac{1}{2}), \quad U = - \frac{\partial f}{\partial \beta}$$

$$\begin{aligned} \rightarrow f &= \frac{1}{V} \ln \prod_{\mathbf{k}, s} \sum_{n_{\mathbf{k}, s}=0,1,\dots} e^{-\beta \hbar \omega_s(\mathbf{k}) (n_{\mathbf{k}, s} + \frac{1}{2})} = \frac{1}{V} \sum_{\mathbf{k}, s} \ln \left(e^{-\beta \hbar \omega_s(\mathbf{k}) \frac{1}{2}} + e^{-\beta \hbar \omega_s(\mathbf{k}) \frac{3}{2}} + \dots \right), \quad x = \beta \hbar \omega_s(\mathbf{k}) \\ &= \frac{1}{V} \sum_{\mathbf{k}, s} \ln e^{-x/2} (1 + e^{-x} + e^{-2x} + \dots) = \frac{1}{V} \sum_{\mathbf{k}, s} \ln \frac{e^{-\beta \hbar \omega_s(\mathbf{k})/2}}{1 - e^{-\beta \hbar \omega_s(\mathbf{k})}} = \\ &= \frac{1}{V} \sum_{\mathbf{k}, s} \left[-\beta \hbar \frac{\omega_s(\mathbf{k})}{2} - \ln (1 - e^{-\beta \hbar \omega_s(\mathbf{k})}) \right] \end{aligned}$$

$$\rightarrow U = \frac{1}{V} \sum_{\mathbf{k}, s} \hbar \omega_s(\mathbf{k}) \left[\frac{1}{2} + \frac{e^{-\beta \hbar \omega_s(\mathbf{k})} (-1)}{1 - e^{-\beta \hbar \omega_s(\mathbf{k})}} \right] = \frac{1}{V} \sum_{\mathbf{k}, s} \hbar \omega_s(\mathbf{k}) (n_s(\mathbf{k}) + \frac{1}{2})$$

$$\rightarrow n_s(\mathbf{k}) = \frac{1}{e^{\beta \hbar \omega_s(\mathbf{k})} - 1} \quad C_V = \frac{\partial U}{\partial T} = \frac{1}{V} \sum_{\mathbf{k}, s} \frac{\partial}{\partial T} \frac{\hbar \omega_s(\mathbf{k})}{e^{\beta \hbar \omega_s(\mathbf{k})} - 1}$$

$$1) \text{ Visoka } T \quad (\beta \rightarrow 0) \quad x = \frac{\hbar \omega}{k_B T} \rightarrow \frac{1}{e^x - 1} = \frac{1}{x + \frac{x^2}{2} + \frac{x^3}{3!}} = \frac{1}{x} \left[1 - \frac{x}{2} + \frac{x^2}{12} \right]$$

$$\rightarrow C_V = \frac{k_B}{V} \sum_{\mathbf{k}, s} 1 - \frac{1}{12} \left(\frac{\hbar \omega_s(\mathbf{k})}{k_B T} \right)^2 = C_{V0} + \Delta C_V$$

$$\text{Dulong-petit: } C_{V0} = k_B \frac{3pN}{V} - \Delta C_V \rightarrow \Delta C_V = \frac{k_B}{12V(k_B T)^2} \sum_{\mathbf{k}, s} (\hbar \omega_s(\mathbf{k}))^2$$

2) Nizka T: $\omega_s(\mathbf{k}) = c_s k$ (pri $T \rightarrow 0$ pride v poštev le linearni del disperzije)

$$C_V = \frac{1}{V} \frac{V}{(2\pi)^3} \frac{\partial}{\partial T} \sum_{\mathbf{k}, s} \int_{\text{1.BZ}} d\mathbf{k} \frac{\hbar c_s k}{e^{\beta \hbar c_s k} - 1} = \frac{3}{2\pi^2} \frac{\partial}{\partial T} \frac{(k_B T)^4}{(\hbar c)^3} \underbrace{\int_0^\infty \frac{x^3 dx}{e^x - 1}}_{\frac{\pi^4}{15}} = \frac{2\pi^2}{15} k_B \left(\frac{k_B T}{\hbar c} \right)^3$$

36. Debyejev približek

- nadomesti vse veje $\omega_s(k)$ z vejami, ki imajo vse enako disperzijo $\omega = ck$

$$N = \frac{4\pi k_D^3}{3 \left(\frac{4\pi}{3}\right)^3} \text{ št. ionov} \rightarrow n = \frac{k_D^3}{6\pi^2}$$

$$C_V = \frac{1}{V} \frac{d}{dT} \sum_{s,k} \frac{\hbar \omega_s(k)}{e^{\beta \hbar \omega_s(k)} - 1} = \frac{1}{(2\pi)^3} 4\pi \cdot 3 \frac{d}{dT} \int_0^{k_D} \frac{k^2 \hbar c k dk}{e^{\beta \hbar c k} - 1}$$

$$\omega_D = k_D c, \quad k_B \Theta_D = \hbar \omega_D$$

$$C_V = \frac{3k_B}{2\pi^2} \left(\frac{\hbar c}{k_B T}\right)^3 \left(\frac{k_B T}{\hbar c}\right)^5 \int_0^{\Theta_D/T} \frac{x^4 e^x dx}{(e^x - 1)^2} = \frac{3k_B k_D^3}{2\pi^2} \left(\frac{k_B T}{\hbar c k_D}\right)^3 \int_0^{\Theta_D/T} \frac{x^4 e^x dx}{(e^x - 1)^2} =$$

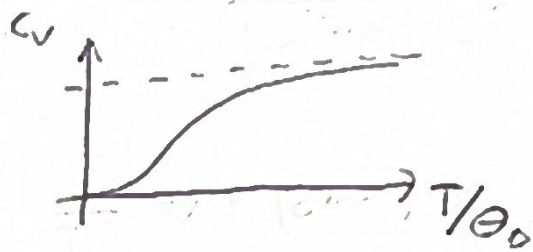
$$= 9nk_B \left(\frac{T}{\Theta_D}\right)^3 \int_0^{\Theta_D/T} \frac{x^4 e^x dx}{(e^x - 1)^2}$$

$T \rightarrow 0$: Rezultat se mora ujemati z realnim

$$C_V = \frac{12\pi^4}{5} nk_B \left(\frac{T}{\Theta_D}\right)^3 = \frac{2\pi^2}{5} k_B \left(\frac{k_B T}{\hbar c}\right)^3 \Rightarrow \frac{1}{c^3} = \frac{1}{3} \sum_s \int \frac{d\Omega}{4\pi} \frac{1}{c_s(k)^3}$$

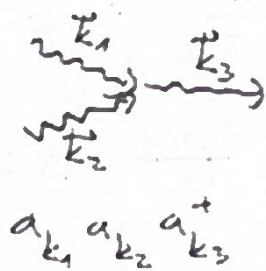
$T \rightarrow \infty$:

$$C_V = 9nk_B \left(\frac{T}{\Theta_D}\right)^3 \int_0^{\Theta_D/T} \frac{x^4}{x^2} dx = 3nk_B \checkmark$$



37. Posledice obstoja anharmonskih členov v razvoju elastične E , term. razt. idealni harm. osc.

- ni interakcij med fononi ($T \rightarrow \infty$)
- ni termičnega raztezanja
- elastične konstante neodv. od p, T
- $C_V(T \rightarrow \infty) = \text{konst.}$



ohranitev E : $\omega_1 + \omega_2 = \omega_3$
 ohranitev GK : $k_1 + k_2 = k_3 + k$

Termično raztezanje:

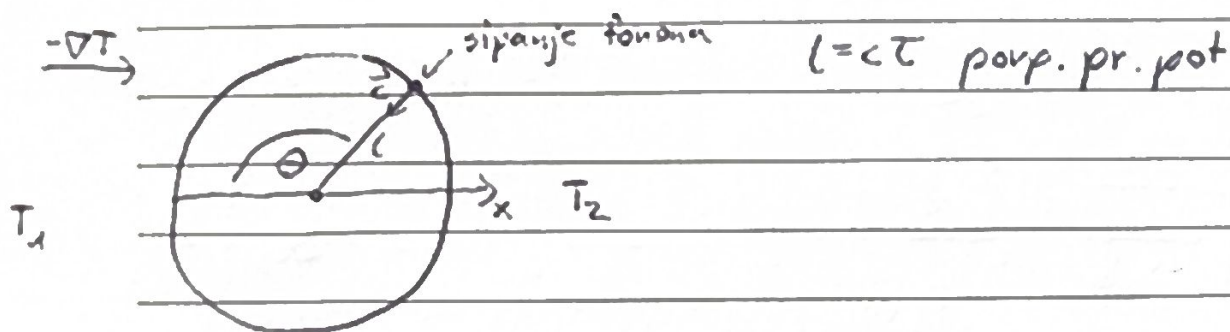
$$U(x) = cx^2 - gx^3 - fx^4, \quad f, g \geq 0. \quad \text{Velja za } cx^2 \gg fx^4, gx^3$$

$$\langle x \rangle = \frac{\int x e^{-\beta U(x)} dx}{\int e^{-\beta U(x)} dx} \approx \frac{\int (x + \beta g x^4) e^{-\beta c x^2} dx}{\int e^{-\beta c x^2} dx} = \frac{(\beta g)(\beta c)^{1/2}}{(\beta c)^{5/2}} = \frac{3}{4} = \frac{3}{4} \frac{g}{c^2} k_B T$$

38. Kinetična teorija mrežne toplotne prevodnosti

Predpostavke:

- Debyejev model, $w = ck$
- fononi se sipljejo, med trki se gibajo s c
- sipanje $\leftrightarrow$ termalizacija $w(x) = w(T(x))$



Čunina nas prispevek sipanja fonona k gostoti en. toka v x.

$$j_{ax} = w(l - x \cos \theta) c_x = \frac{1}{4\pi} \int_0^{2\pi} d\varphi \int_0^\pi \sin \theta d\theta \underbrace{c \cos \theta}_{c_v} w(l - x \cos \theta) \stackrel{\cos \theta = x}{=} \\ = \frac{1}{2} \int_{-1}^1 x c(w(x) - l x \frac{dw}{dx} \Big|_x) dx = \frac{1}{3} c l \left(-\frac{dw}{dx} \right)$$

$$\frac{dw}{dx} = \frac{dw}{dT} \frac{dT}{dx} = c_v \frac{dT}{dx} \rightarrow j_{ax} = \frac{1}{3} c l c_v (-\nabla T) \rightarrow \kappa = \frac{1}{3} c l c_v$$

39. Ocena T odvisnosti c_v in umklapp procesi

$$c_v(T) \propto \begin{cases} T^3, & T \rightarrow 0 \\ 3nk_B T, & T \rightarrow \infty \end{cases}$$

$T \rightarrow 0$:

• normalni procesi

$$T \rightarrow \infty \Rightarrow \tau \propto \frac{1}{\omega}$$

$$n_i = \frac{1}{e^{\beta \hbar \omega_i} - 1} \approx \frac{k_B T}{\hbar \omega_i}$$



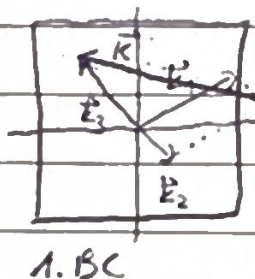
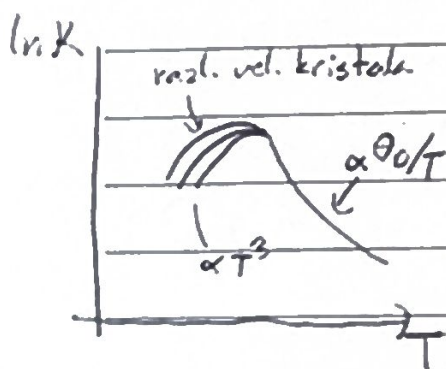
GK se ohranja

$$\sum_i \hbar \mathbf{k}_i n_i = \sum_j \hbar \mathbf{k}_j n_j$$

PREČ PO

$$\rightarrow \tau \propto T^{-x}, x \in [1, 2]$$

• Umklapp procesi



$$\sum_i \hbar \mathbf{k}_i n_i = \sum_j \hbar \mathbf{k}_j n_j + \hbar \mathbf{K}$$

$$n_i = \frac{1}{e^{\beta \hbar \omega_i} - 1} \approx e^{-\theta_0/T}$$

$$\rightarrow \tau \propto e^{\theta_0/T}$$